SOFTWARE REQUIREMENTS SPECIFICATION

RealmForge

Team 20
Aeolus Software

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Revision 0: October 15, 2026

Revision History

Name	Date	Reason For Changes	Version
Andy Huynh	2026-10-05	First Draft	1.0
Andreas Sideris	2026-10-10	P3 and P4 drafts	1.0
Andreas Sideris	2026-10-12	E3 draft	1.0
Andreas Sideris	2026-10-13	E2 draft	1.0
Andreas Sideris	2026-10-15	AI disclosure and Reflection updated	1.0

This document follows the requirements documentation structure presented in the Handbook of requirements and business analysis, by Bertrand Meyer.

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Comment: *Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders.*

G.1 Context and overall objective

Comment: *High-level view of the project: organizational context and reason for building a system.*
Comment: *This chapter should not be empty!*

Goal 1. This is a goal example. If you need explicit (and automatic) numbering, you can use the definitions in the `.tex` template. Is is refined by 3

Requirement 2. *This is a requirement example. It illustrates how numbering is continuous and cross-types (if this is what you need).*

G.2 Current situation

Comment: *Current state of processes to be addressed by the project and the resulting system.*

Requirement 3. *This is a requirement example. It refines 1*

G.3 Expected benefits

Comment: *New processes, or improvement to existing processes, made possible by the projectâ€™s results.* **Comment:** *This chapter should not be empty!*

G.4 Functionality overview

Comment: *Overview of the functions (behavior) of the system. Principal properties only (details are in the System book).*

Setting properties: The system application starts up and the the user is shown options that will allow the user to pick and change various settings during the start of the application, so that the user can experiment with different variants of the game. Comment: *Less detail on the button pressing*

Traversal: After the player has changed some properties or just simply pressed the play option menu button, the game begins and the system displays the HUD as well other UI components such as the main game screen and travel options. The player can navigate to another area by pressing the up, down, left, or right menu option buttons go to another area indicated by the change in the background scenery.

Comment: *Branching patchs, endpoints, you need to win*

Combat: The player enters combat randomly when traversing through areas. Once in combat, the battle screen is shown as well as the enemy the player has to defeat. The player is shown options to attack or escape during the encounter. If the player chooses to attack the enemy, an attack action is displayed followed by the enemies opposing attack. If the player defeats the enemy, the player is rewarded with resources at the end. Comment: *You win or die*

Returning to the Hub: If the player escapes, then there is no reward and the player goes back to the main game screen. The gameplay loop of traveling and encounter enemies continues until the player returns to the starting zone (the central hub), with the return option or when the player is defeated. If the player returns to the starting zone, the player is allowed to spend the resources accumulated throughout the journey to construct buildings and upgrades that will enhance the experience of the next gameplay loop. Comment: *Change to only comming back when you die* Comment: *Immediately enter combat* Comment: *Remove randomization* Comment: *You know what your getting into, but not complete knowledge* Comment: *Higher risk, higher reward*

End of game: Once the player finishes spending resources, the player may return to the main gameplay loop of traveling and fighting enemies to gain resources. If the player is defeated, the game ends and player is returned to the start of the game where the player chooses options for changing parameters.

G.5 High-level usage scenarios

Use Case 001 (UC-001): Selecting Properties

1. User launches the RealmForge software.
2. User selects feature variation configurations.

3. User decides which features of the game to be changed.
4. User finishes setting up by beginning the game.

Use Case 002 (UC-002): Playing a full session

1. User selects to begin a new game session.
2. User enters the main game world.
3. User decides which area of the game world to go to explore based on the information given for each area.
4. User selects which area to go to and begins traveling.
5. User enters a new area and begins an encounter with an enemie/enemies (see UC-003).
6. User is successful (continue below) or fails in the encounter (see UC-005).
7. User continues traveling the game world (repeats step 2 through 6).
8. User completes the win condition for the game session (see UC-005).

Use Case 003 (UC-003): Battling enemies

1. User enters an encounter with the enemie/enemies.
2. User selects attacks to defeat the enemie/enemies.
3. User repeats step 2 until the enemie/enemies are defeated.
4. User collects rewards for successfully completing the encounter.
5. User returns to game's world map.

Use Case 004 (UC-004): Upgrading and purchasing items

1. User enters town/city/central hub.
2. User chooses and selects an available option(s) for purchasing.
3. User confirms the purchase.
4. User repeats steps 2 and 3 until the user is satisfied or resources are spent.
5. User leaves the town/city/central hub to continue the game session.

Use Case 005 (UC-005): Continuing or start a new game

1. User completes a game session or fails in between (see UC-002).

2. User is given an option to return to the central hub (see UC-004) or quit the current game session (see below).

3. User selects to change features (see UC-001) or proceeds with the same configurations.

4. User begins a new game.

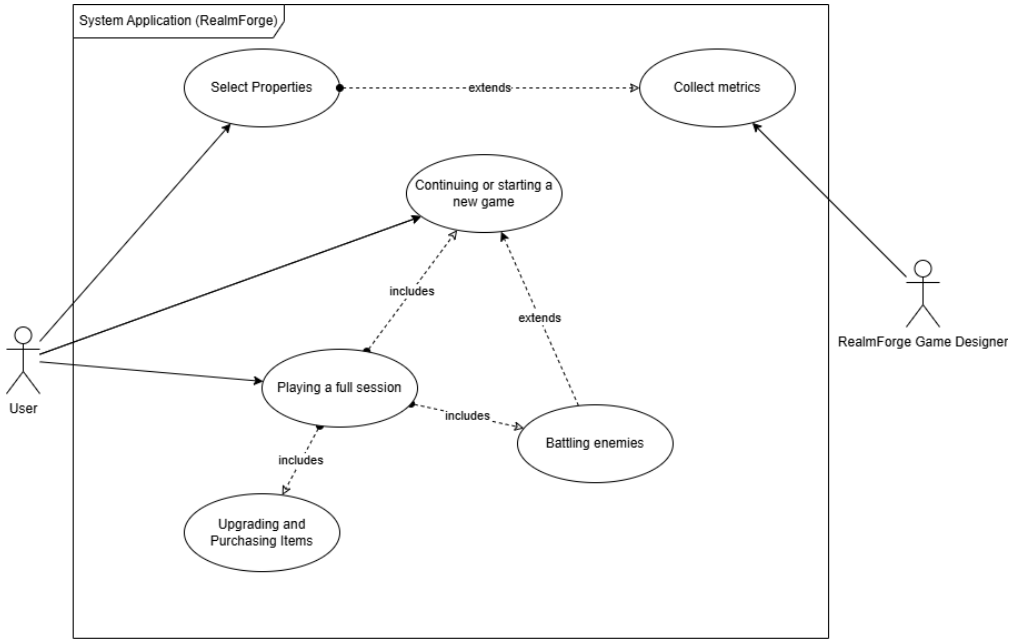


Figure 1: Use Case Diagram

G.6 Limitations and exclusions

Comment: *Aspects that the system need not address.*

G.7 Stakeholders and requirements sources

Comment: *Groups of people who can affect the project or be affected by it, and other places to consider for information about the project and system.* **Comment:** *This chapter should not be empty!*

Environment

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Comment: *The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate.*

E.1 Glossary

Comment: *Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. This chapter should not be empty!*

- **User** - The main stakeholder that is operating the system; providing inputs and receiving outputs.
- **Player** - Is another name refering the user.
- **User-Interface (UI)** - The entire set of displayed service elements the player used to navigate the system.
- **Heads-Up-Display (HUD)** - The set of interface elements displayed ontop of the actually gameplay that contains vital information helpful to the player or user.
- **Central Processing Unit (CPU)** - The central processor for a computer.
- **Streaming Single Instruction, Multiple Data Extentions 4 (SSE4.2)** - A processor instruction set used by most modern processors.
- **Advanced Micro Devices (AMD)** - An American semiconductor company that produces processors.
- **Graphics Processing Unit (GPU)** - A computer's processor for displayed graphics.

- **Open Graphics Library (OpenGL)** - An application programming interface used to render computer graphics.
- **NVIDIA Corporation** - An American technology company that produces graphics processors.
- **Graphics Core Next (GCN)** - A series of graphics processor architectures that is used in AMD graphics processors.
- **Random Access Memory (RAM)** - Short-term memory used by a computer's processor.
- **Frames per Second (FPS)** - The number of graphical frames displayed by a computer program per second.
- **Personal Computer (PC)** - A computer owned by individuals, for personal use.
- **Verification and Validation (VnV)** - The process of ensuring that a design satisfies its requirements, and its requirements are aligned with the design's purpose (generally).
- **Feature-Oriented Domain Analysis (FODA)** - A domain analysis method, focusing on features of a design and their variations.
- **Proof of Concept (POC)** - An initial version of a design, intended to demonstrate the possibility of its implementation without being a full prototype.
- **Inventory** - The set of resources, items, and currency store within code and available to access for the player

E.2 Components

Included below is a System Context Diagram, outlining at a high level how the system will interact with external components:

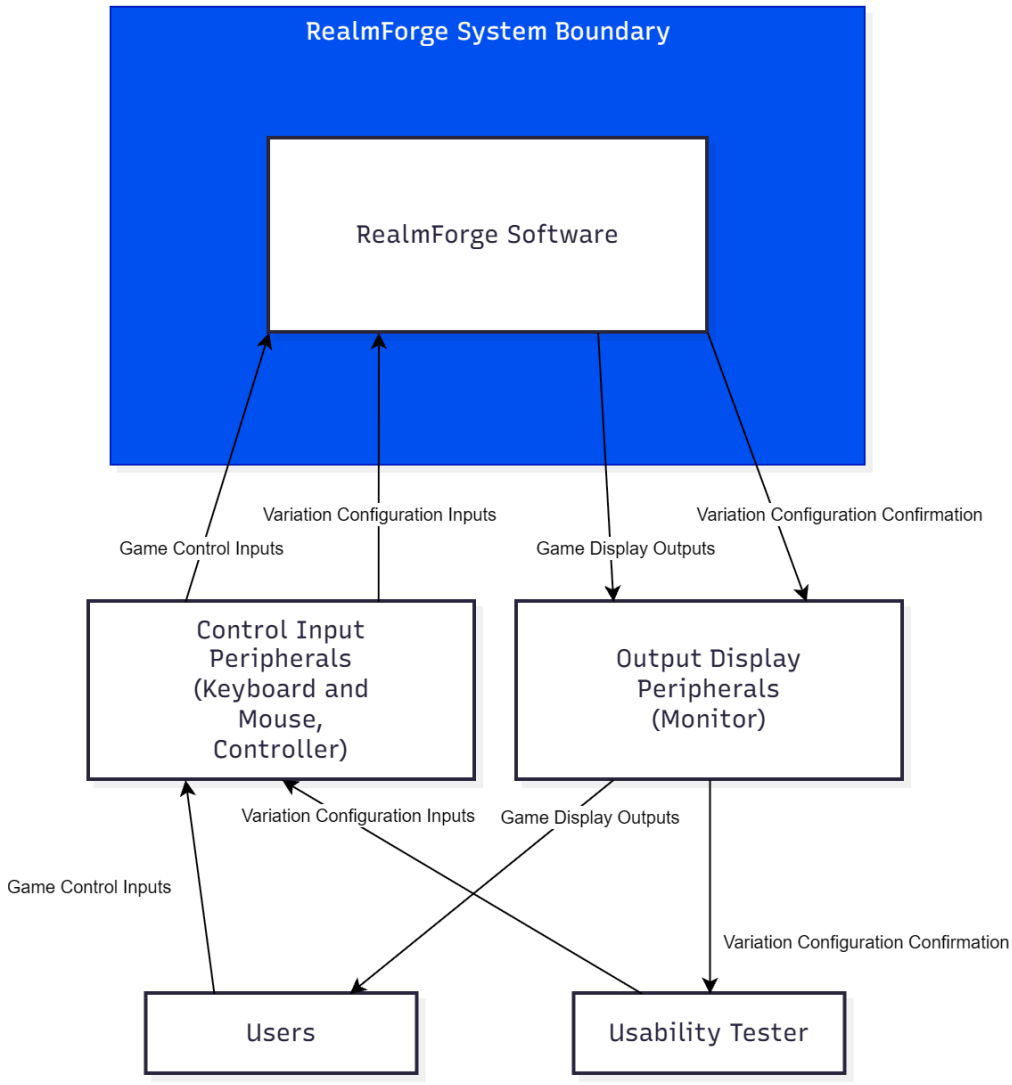


Figure 2: RealmForge System Context Diagram

The RealmForge software (also denoted as the system) will require the use of the following external components to operate successfully:

- **Control Input Peripherals:** Devices used to provide input to the system. This could include a keyboard and mouse, or a controller used to provide input. The system will require inputs from these devices to affect the state of the video game it

executes (during gameplay), and to configure the system's feature variation points during testing.

- **Output Display Peripherals:** A device used to display outputs provided by the system. This could be fulfilled by a monitor, or another display device (such as a television). The system will require this device to display graphical outputs during gameplay (video game graphics), as well as confirmation of feature variation configurations (variation config. menu graphics).

E.3 Constraints

ENV-001: The system shall be compatible with user machines running Windows 10 or 11 (64-bit) operating systems that meet the following recommended specifications:

- **CPU:** x86_64 CPU with SSE4.2 support, with 4 physical cores or more. Example: Intel Core i5-6600K, AMD Ryzen 5 1600.
- **GPU:** Dedicated graphics GPU with full OpenGL 4.6 support. Example: NVIDIA GeForce GTX 650, AMD Radeon HD 7750 (GCN 1.0).
- **RAM:** A minimum of 8 GB of RAM.

Rationale: This environment constraint is imposed to focus development of RealmForge on the most commonly used operating system for PC video games, Windows. The system may be extended to support other operating systems in the future if possible. The recommended specifications are included to ensure user machines can fully execute the system, accompanying any future decisions to extend its capabilities.

Fit Criterion: The system can be compiled and executed on machines running Windows 10 or 11 (64-bit) operating systems, meeting the recommended specifications outlined above, at a stable 60 fps.

Traceability: N/A

ENV-002: The system shall provide user input capabilities for keyboard and mouse peripherals, as well as common controllers (PlayStation, Xbox).

Rationale: The system must accommodate the use of various input methods (keyboard and mouse, different controllers) to facilitate usability studies incorporating these varying input methods.

Fit Criterion: The system can receive all game inputs required for its successful operation from the following input methods: USB Keyboard and Mouse, Xbox Series X/S, Xbox One, or Xbox 360 controllers, and PlayStation DualSense or DualShock 4 controllers.

Traceability: N/A

ENV-003: The system shall provide at least one explicit feature variation capability, that can be selected without recompiling the system's executable file.

Rationale: This constraint is imposed by the purpose of the development project (outlined in the Problem Statement document). The system must provide explicit feature variation capabilities to facilitate usability testing that will focus on the effectiveness of different variations.

Fit Criterion: The system provides at least one explicit feature variation capability, that can be selected and managed at run time, while the system is executing.

Traceability: N/A

E.4 Assumptions

Comment: *Properties of the environment that may be assumed, with the goal of facilitating the project and simplifying the system.*

E.5 Effects

Comment: *Elements and properties of the environment that the system will affect.*

E.6 Invariants

Comment: *Properties of the environment that the systemâ€™s operation must preserve.*

System

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Comment: *The System book refines the Goal one by focusing on more detailed requirements about the system under development, mainly its constituents, behaviors and properties.*

S.1 Components

Comment: *Overall structure expressed by the list of major software and, if applicable, hardware parts.*
Comment: *This chapter should not be empty!*

S.2 Functionality

Comment: *One section, S.2.n, for each of the components identified in S.2, describing the corresponding behaviors (functional and non-functional properties).* **Comment:** *This chapter should not be empty!*

S.3 Interfaces

Comment: *How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs).*

S.4 Detailed usage scenarios

Comment: *Examples of interaction between the environment (or human users) and the system: use cases, user stories.*

S.5 Prioritization

Comment: *Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality.*

S.6 Verification and acceptance criteria

Comment: *Specification of the conditions under which an implementation will be deemed satisfactory.*

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Comment: *The Project book describes all the constraints and expectations not about the system itself, but about how to develop and produce it.*

P.1 Roles and personnel

Comment: *Main responsibilities in the project; required project staff and their needed qualifications.*

P.2 Imposed technical choices

Comment: *Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters.*

P.3 Schedule and milestones

The project's primary schedule will follow the milestones and dates below, each named according to the deliverables they must produce (as discussed in the project's development plan):

- M1 Problem Statement, Development Plan, PoC Plan, and Team Charter (Due by September 28, 2026)
- M2 SRS and Hazard Analysis Revision 0 (Due by October 19, 2026)
- M3 VnV Plan Revision 0 (Due by November 2, 2026)
- M4 Design Document Revision -1 and CD-Fall (FODA) (Due by November 16, 2026)

- M5 PoC Demo (Due by November 23, 2026)
- M6 Design Documentation Revision 0 (Due by January 18, 2027)
- M7 Revision 0 Demo (Due by February 1, 2027)
- M8 VnV Report Revision 0 and CD-Winter Revision 0 (Usability Report) (Due by March 8, 2027)
- M9 Final Demonstration Revision 1, EXPO Demonstration (Due by March 15, 2027)
- M10 Final Documentation (Revision 1) (Due by March 31, 2027)

P.4 Tasks and deliverables

The tasks required to be completed to develop RealmForge (and its feature variation capabilities) will be managed using an iterative and incremental approach. Within each milestone (in this context the milestones associated with source code development), team members will divide the work to be done into suitable issues, viewable in RealmForge's GitHub Project. The issues will be coordinated using a Kanban method, tracking their progress and review status. The software features associated with these issues will be regularly planned, developed, reviewed, tested, and deployed using a feature branch workflow, contributing towards each set milestone.

A schedule of development deliverables pertaining to the milestones listed above is available below, complete with relevant requirement sets that are planned to be implemented at the completion of each deliverable. Milestones that purely involve documentation development are omitted, as this schedule is intended to outline the completion of software development tasks and their associated requirements.

Detailed Schedule:

- M5 PoC Demo
  - **Deadline:** November 23, 2026
  - **Software Deliverables:** RealmForge Proof of Concept Build
    - **Associated Software Tasks:**
    - Review PoC plan and requirements
    - Develop PoC features
    - Review implemented PoC features on a recurring basis during development
    - Test PoC features
    - Verify associated requirements (outlined below)

        * Deploy PoC
    - **High Priority Requirements:** NEED TO ADD
    - **Requirement Sets Implemented:** NEED TO ADD

- M7 Revision 0 Demo
    - **Deadline:** February 1, 2027
    - **Software Deliverables:** RealmForge Revision 0 Build
        * **Associated Software Tasks:**
        * Review Revision 0 design documentation and requirements
        * Develop Revision 0 features
        * Review implemented Revision 0 features on a recurring basis during development
        * Test Revision 0 features
        * Verify associated requirements (outlined below)
        * Deploy Revision 0
    - **High Priority Requirements:** NEED TO ADD
    - **Requirement Sets Implemented:** NEED TO ADD

- M9 Final Demonstration Revision 1, EXPO Demonstration
    - **Deadline:** March 15, 2027
    - **Software Deliverables:** RealmForge Revision 1 Build
        * **Associated Software Tasks:**
        * Review Revision 1 design documentation (in progress) and requirements
        * Develop Revision 1 features
        * Review implemented Revision 1 features on a recurring basis during development
        * Test Revision 1 features
        * Verify associated requirements (outlined below)
        * Deploy Revision 1
    - **High Priority Requirements:** NEED TO ADD
    - **Requirement Sets Implemented:** NEED TO ADD

P.5 Required technology elements

Comment: *External systems, hardware and software, expected to be necessary for building the system.*
The application will be run, developed and tested using hardware that supports the Window's OS. Other OS's such as Apple's or other platforms will be considered after the initial development showcase. Inputs are handled either with a standard keyboard and mouse or with a connected controller using standard button layout (analog sticks, d-pads, shoulder and 4-buttons layout). The controller should be compatible with any Windows OS, similarly to the mouse and keyboard. No unique gaming peripheral is necessary. No enhanced audio peripheral is necessary. The game output will be displayed using any standard monitor connected to the computer hardware.

The system will be built using the Godot game engine including using Godot's own coding language, GDScript. The IDE will be up to each individual team member's preference. GDUnit4 and GDLint and other external third-party modules will be used for CI/CD as a safeguard for good coding practices and coverage. The project will use Github's version control and cloud storage for the source code and documents. The GDLint and GdUnit4 unit tests and linting will be implemented using using Github Actions available in the Github website; this is so that errors and bugs will be caught and reiterated early on. Latex will be used for writing important documents such Hazard Analysis, Development etc. Issues pertaining to the project will be raised on the Github website and written/handled with its own built-in issue tracker. Other important project management tasks will be written using Github Project's Kanban board for organization and easy tracking. Microsoft teams is the main technology platform that is used for communication such as in group chats and voice-calls.

TODO: Add glossaries for OS, CI/CD, IDE, and discuss with the team about issue tracker with trello or github.

P.6 Risks and mitigation analysis

Comment: *Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise.*

P.7 Requirements process and report

Comment: *Initially, description of what the requirements process will be; later, report on its steps.*

Appendix

Appendix — Generative AI Use Disclosure

AI Tools Used

- OpenAI GPT-6
- Google Search AI Overview (Google Gemini)

How and Where Used

AI tools were used for assistance with LaTeX syntax and commands, during the development of sections E2, E3, S4, P3, and P4, and when developing LaTeX formatting for the document as a whole. Specifically, AI tools were used for assistance in creating the LaTeX custom requirement format macro used throughout the document, as well as resolving various minor formatting and table of contents bugs. Additionally, these tools were also used in these sections to conduct brief 'sanity checks' of content in accordance with Meyer's requirements document template (the template description was provided). These 'sanity checks' did not substitute full reviews of the sections, and were more used as a brief second opinion. In these sections, AI tools were not used to generate any text or wholly generate any of the ideas or conclusions presented.

Verification of AI-Generated Content

Any LaTeX syntax or code provided by AI tools was verified by testing compilation in LaTeX documents, and ensuring the resulting output was correctly formatted and matched the needs of the document. Conclusions from brief 'sanity checks' of the above sections using AI tools were independently verified for correctness using course materials and documentation of Meyer's requirements template, and no ideas or text were newly generated by AI tools and presented in the document.

Appendix — Reflection

Reflection Questions Answered:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

Andreas:

1. When writing this deliverable, adopting Bertrand Meyer's requirement template and completing its sections was helpful and applicable to our project, since all team members were familiar with the template previously, and it is provided with thorough documentation. Understanding the template previously, and being able to review its full documentation helped our team to understand what was required to complete each section. Additionally, writing this deliverable forced our team to deeply consider various details of the project and requirements that the system must satisfy, which helped to enable us to make important project decisions. Obviously one must decide on various project details when writing a requirement specification, but completing this SRS helped us to solidify and exactly outline the outputs needed to be produced by the system, and other constraints it must follow.
2. One pain point I experienced during this deliverable was coordinating the many sections to be written and decisions to be made as a group. Ideally, all project and system planning decisions will be made together, but work must be divided at some point so that each team member can asynchronously complete their tasks (while reviewing and discussing with the team when needed). This at times led to more

administrative overhead work, as various sections needed to be iteratively reviewed and discussed to ensure that the document did not contain major contradictions. However, this coordination work was required for the success of the document, and was lessened in certain scenarios by initially discussing clear expectations for various sections as a group, to ensure that the team member completing a specific section had the information required to do so correctly.

3. A majority of the sections I contributed to did not involve the production of formal requirements, except E.3 (Constraints), which included formal environmental constraint requirements. Some of these requirements (feature variation points, and user input device requirements) were primarily influenced by discussions with our TA and professor, using their recommendations on including feature/system variability in the project, and specifically doing so by having players use different input methods (controller, keyboard and mouse). Additionally, the environment requirement included concerning the operating system RealmForge will support was influenced by discussions with peers that could be included in our user groups (those who play video games and are familiar the roguelite genre, those who are not, etc.). It was found (as expected) that most users tend to play PC video games on machines running modern Windows operating systems, as they are most commonly available and easy to configure to play popular video games.

4. Of the courses I have taken (and are currently taking), SFWRENG 3RA3 (Requirements) and 4HC3 (Human-Computer Interaction) are particularly helpful to our team (and myself specifically) for our capstone project. SFWRENG 3RA3 introduced all team members to the use of Meyer's requirements template and principles of requirements engineering as a whole, which helped us to complete this document, and construct requirements for the system. SFWRENG 4HC3 provides our team with important knowledge and techniques for effective user interface and experience design, which will be particularly helpful for our project, since it primarily focuses on interface design and usability testing.

5. One skill the team will collectively need to acquire (or further develop) is video game development. This will include becoming more familiar with our chosen game development engine and language, as well as better understanding game development principles. This will be especially important for myself, as I have limited experience in programming for video game development. Another area of knowledge that will need to be further developed for each team member concerns usability testing techniques and principles, as each of us are just beginning to study these concepts in SFWRENG 4HC3 (as discussed above).

6. For improving game development skills, team members could use provided game engine tutorials and documentation for the game engine used, review YouTube and web article lessons on game development principles, or read books that provide instruction on game development principles. I will individually aim to primarily follow tutorials and documentation for our chosen game engine, since the initial

technical skills of game development are new to me. I will also aim to improve my understanding of game development principles through various tutorials and lessons when possible. For developing interface design and usability testing knowledge, team members could take notes and review how to best apply the concepts taught in SFWRENG 4HC3 to our Capstone project specifically, or consult other standards or tutorial sources for interface design and usability testing. I will primarily aim to review and understand how the concepts taught in 4HC3 could be applied to our project, as I currently am developing my understanding of these concepts well in the context of the course, and believe they could be effective when applied to the development of RealmForge.

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Christopher:

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Andy:

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